

Effectiveness of Sitagliptin in Indian Patients with T2DM

Real-world Retrospective EMR Based Study

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Introduction

- According to the 2021 ICMR-INDIAB-17 national cross-sectional study, diabetes prevalence in India reached 11.4%, with an additional 15.3% of the population diagnosed with prediabetes,¹ notably type 2 diabetes mellitus (T2DM) being the most prevalent form, accounting for 90-95% of all diabetes cases.²
- In diabetic patients there is high prevalence of being overweight and obese with some studies reporting the prevalence to be as high as 80-90%.^{3,4}
- RSSDI 2022 Guidelines recommends a patient-centric and evidence-based approach while managing diabetes. Treatment decisions should be based on glycaemic control and safety profile including hypoglycaemia risk, comorbidities, route of administration, patient preference, and treatment costs in Indian patients with T2DM.⁵
- Multiple oral anti-diabetic class of drugs available and DPP-4 inhibitors are effective in improving glycaemic both as monotherapy and as add-on to metformin.⁶⁻⁸
- Sitagliptin, a dipeptidyl peptidase 4 (DPP-4) inhibitor, inhibits the DPP-4 enzyme and deactivates incretin hormones such as glucagon like peptide-1 (GLP-1). It stimulates insulin secretion in a glucose-dependent manner, suppresses glucagon synthesis, lowers hepatic gluconeogenesis, and slows gastric emptying. It has beta cell preserving action, is weight neutral and safe in patients with established CVD.⁶⁻⁹
- Sitagliptin is a useful treatment option for patients with T2DM, as it improves glycaemic control with a low incidence of hypoglycaemia and less adverse effects.¹⁰
- Real-world evidence from retrospective studies based on electronic medical records (EMR), offers a comprehensive view of treatment outcomes in different patient groups.^{11,12}

Objectives

- A real-world, retrospective, observational, EMR-based study was conducted to understand the effectiveness of sitagliptin and sitagliptin + metformin FDC in Indian patients with T2DM. The study was approved by Independent Ethics Committee and registered on CTRI (CTRI/2023/10/058366).

Materials and Methods

- The EMR data of adults aged ≥ 18 years of either gender having T2DM, between 2017 and 2023 who were prescribed with sitagliptin or sitagliptin + metformin FDC, with or without other OHAs, with at-least 1 follow-up visit at 3 months was retrieved. The data here presents a *post-hoc* analysis of treatment-total pool of patients who received sitagliptin and assessed the effectiveness of sitagliptin on glycosylated haemoglobin (HbA1c), fasting (FBG) and postprandial blood glucose (PPBG) levels from baseline to 3 months with HbA1c $\geq 7\%$ at baseline.
- Primary end-points**
 - Mean change in HbA1c from baseline to 3 months in patients with HbA1c $\geq 7\%$ at baseline
- Secondary end-points**
 - Mean change in PPBG and FBG from baseline to 3 months in patients with baseline HbA1c $\geq 7\%$
 - Proportion of patients achieving HbA1c $< 7\%$

Results

In the current EMR data, 2,794 patients were prescribed with sitagliptin alone of which 1,008 had HbA1c $\geq 7\%$ at baseline.

Mean change in HbA1c from baseline to 3 months

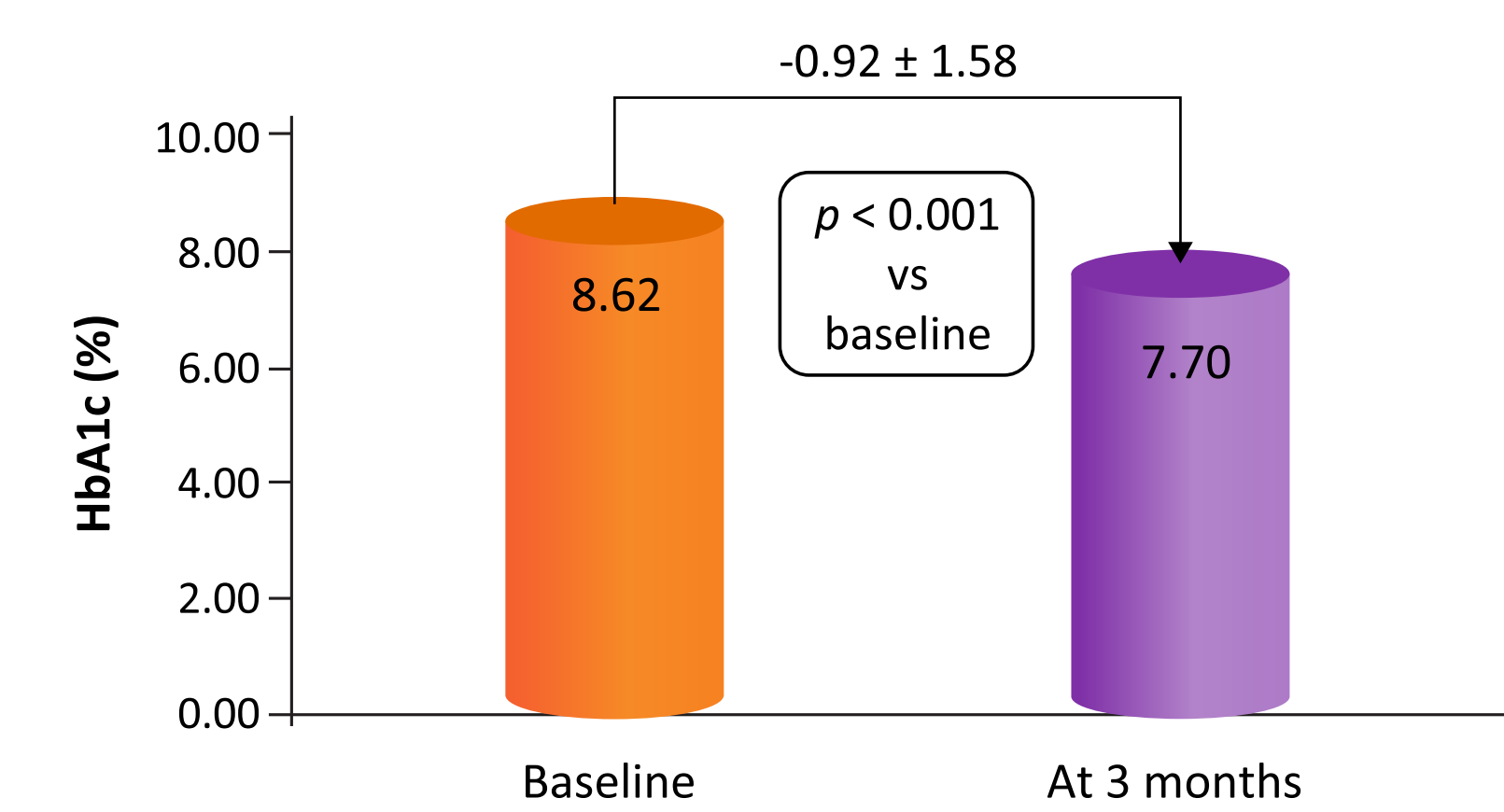


Figure 1: Mean change in HbA1c from baseline to 3 months (N = 1,008)

A statistically significant reduction in HbA1c was observed from baseline to 3 months [(8.62 $\pm$ 1.46 to 7.70 $\pm$ 1.16, $p < 0.001$, N = 1,008) Figure 1].

Mean change in FBG from baseline to 3 months

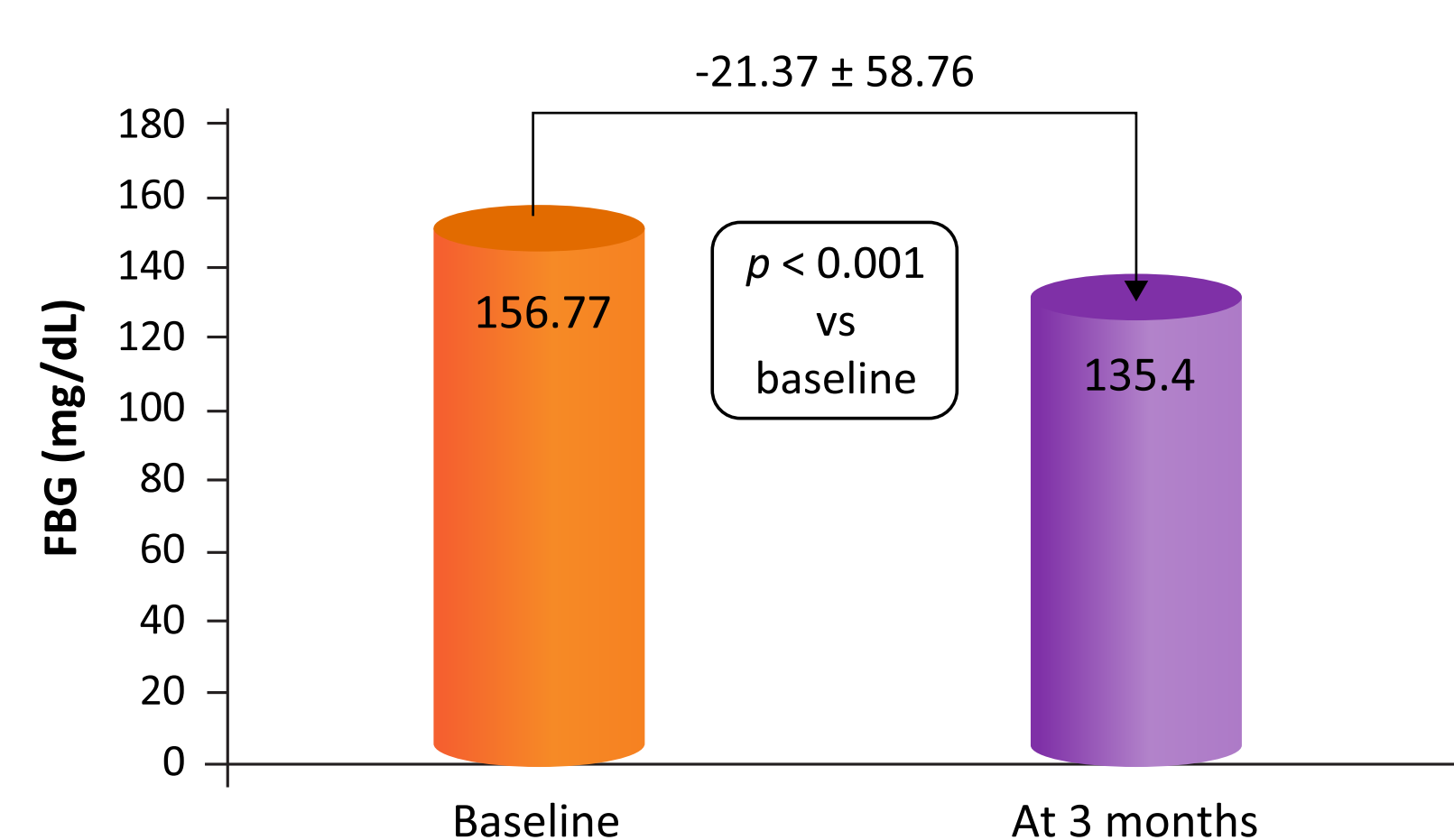


Figure 2: Mean change in FBG from baseline to 3 months (N = 914)

A statistically significant reduction in FBG was observed from baseline to 3 months [(156.77 $\pm$ 53.63 to 135.40 $\pm$ 38.98, $p < 0.001$, N = 914) Figure 2].

Mean change in PPBG from baseline to 3 months

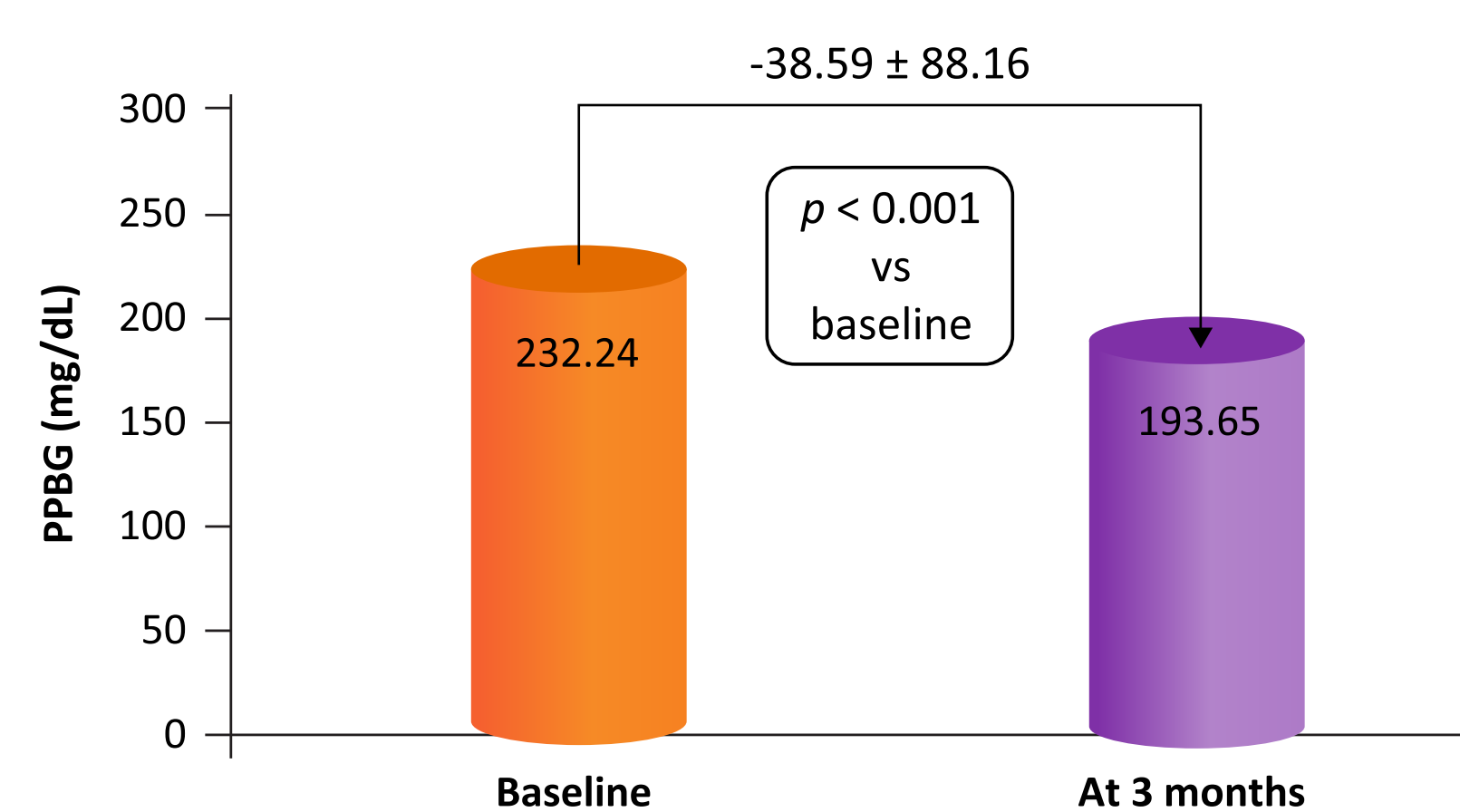


Figure 3: Mean change in PPBG from baseline to 3 months (N = 790)

A statistically significant reduction in PPBG was observed from baseline to 3 months [(232.24 $\pm$ 81.71 to 193.65 $\pm$ 61.77, $p < 0.001$, N = 790) Figure 3].

Proportion of patients achieving HbA1c < 7%

Proportion of patients achieving HbA1c $< 7\%$ at 3 months was 25.79% (N = 260/1008).

There was statistically significant change in the HbA1c, FBG and PPG at the end of 3 months ($p < 0.001$).

Limitation

- This was a single arm analysis of EMR data captured on digital platform and the database that we used, does not capture safety data

Conclusion

- Sitagliptin monotherapy demonstrated rapid and effective glycaemic control regardless of demographic or clinical characteristics in real-world study. The beta cell preservation action and potential of having low risk of hypoglycaemia, makes sitagliptin an effective oral diabetic agent in Indian diabetic patients.
- Sitagliptin significantly reduced FPG and PPBG crucial for comprehensive diabetes management.
- Sitagliptin effectively helped a notable proportion of patients to achieve HbA1c levels below 7%, supporting its use in real-world clinical practice for better overall management of T2DM.
- Long term studies are needed to assess effectiveness and safety of sitagliptin.

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Abbreviations: CVD, cardiovascular disease; EMR, electronic medical records; FBG, fasting blood glucose; FDC, fixed-dose combination; HbA1c, glycosylated haemoglobin; ICMR-INDIAB, Indian Council of Medical Research-India Diabetes; DPP4, dipeptidyl peptidase 4; OHAs, oral hypoglycemic agents; PPBG, postprandial blood glucose; RSSDI, Research Society for the Study of Diabetes in India; T2DM, type 2 diabetes mellitus

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